

Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)(i)	force = $(6.67 \times 10^{-11} \times 4.0 \times 10^{30} \times 2.0 \times 10^{30}) / (3.3 \times 10^{12})^2$	C1
	= 4.9×10^{25} N	A1
1(b)(ii)	$a = F / m = (4.9 \times 10^{25}) / (4.0 \times 10^{30})$ = $1.2 \times 10^{-5} \text{ m s}^{-2}$	A1
1(b)(iii)	$a = r\omega^2$ and $\omega = 2\pi / T$ or $a = v^2 / r$ and $v = (2\pi r / T)$	C1
	$r = (3.3 \times 10^{12}) / 3$	C1
	$T = 2\pi [(3.3 \times 10^{12}) / (3 \times 1.2 \times 10^{-5})]^{1/2}$ = $1.9 \times 10^9 \text{ s}$	A1
1(b)(iv)	acceleration: B greater than A	B1
	linear speed: B greater than A	B1
	period: B equal to A	B1

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